

EmilyKin Hybrid Metagenomics Pipeline

5 samples (CAN 1-5) · Nanopore ONT + Illumina NovaSeq · Snakemake workflow

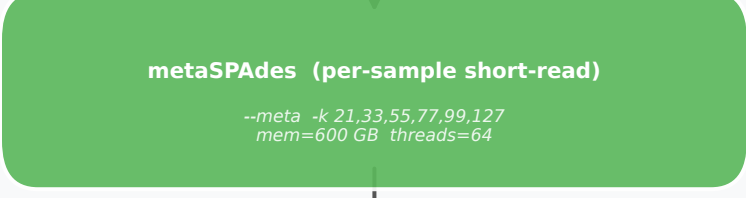
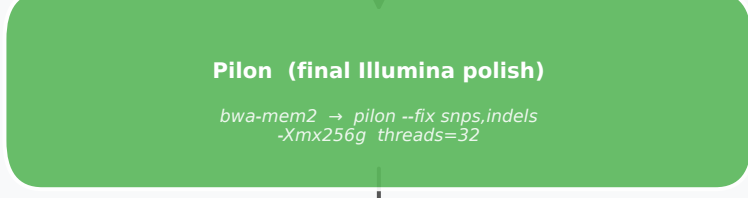
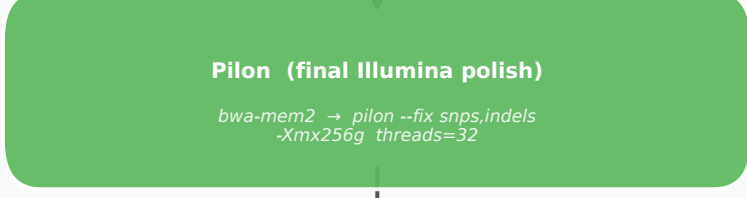
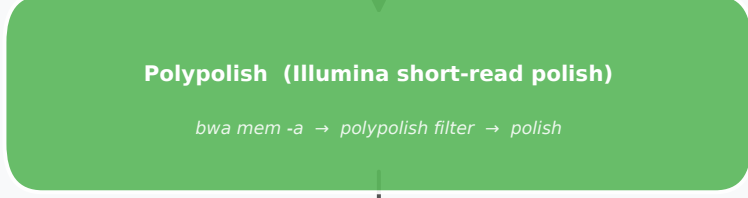
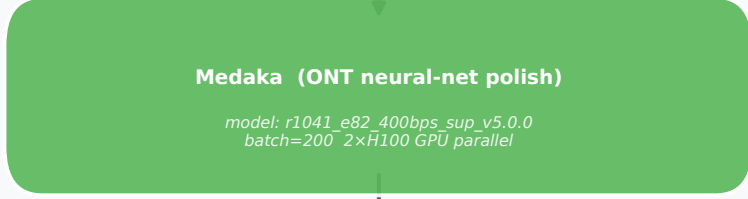
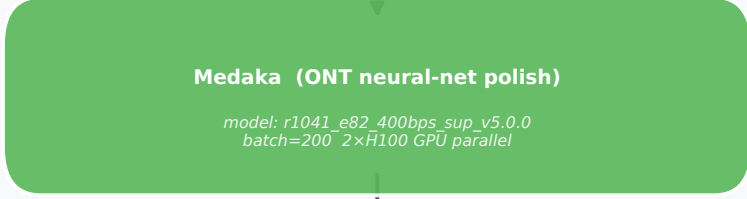
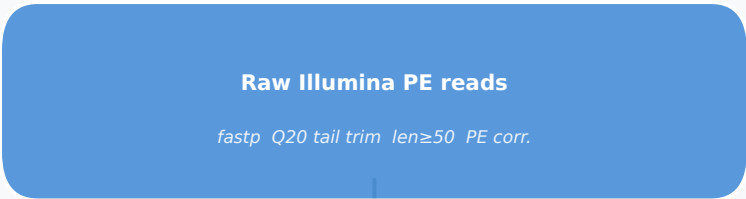
Per-Sample Hybrid (×5 samples)

Hybrid Co-Assembly (all 5 concatenated)

Per-Sample Short (×5, metaSPAdes)

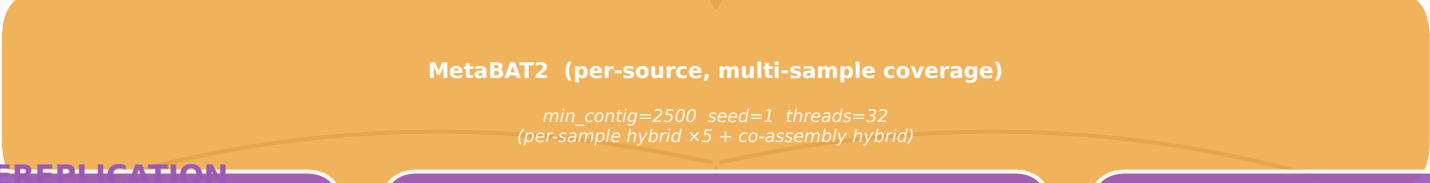


*fastp clean reads
→ per-sample + co-assembly*



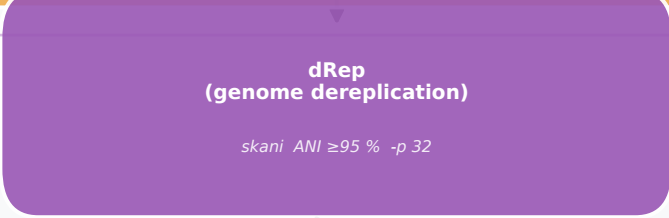
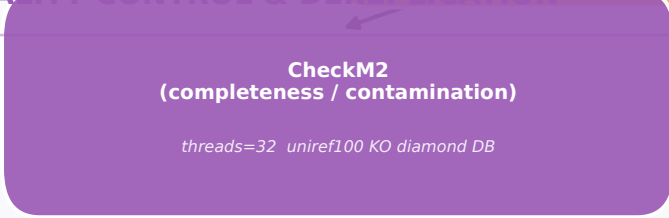
▼ Final assemblies per source ▼

BINNING (MetaBAT2 — hybrid assemblies + co-assembly)



Bins from all
sources pooled
before QC

MAG QUALITY CONTROL & DEREPLICATION

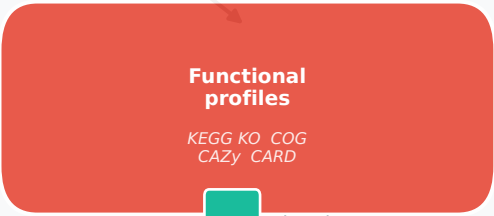
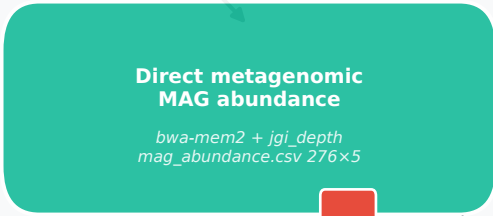
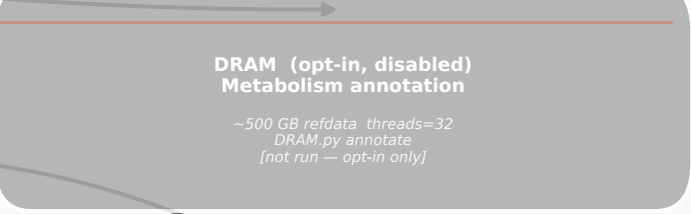
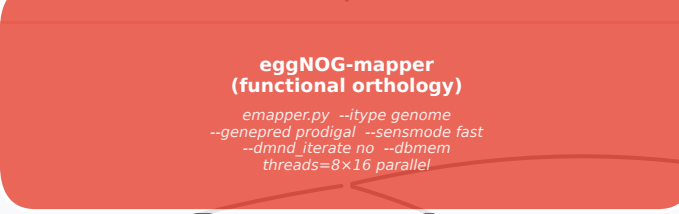
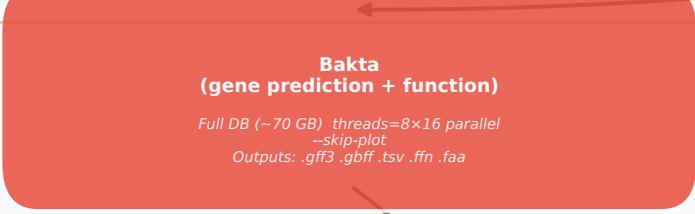


Filter: completeness ≥50%
contamination ≤10%

276 dereplicated MAGs
(species-level clusters)

NCBI-compatible lineage
assigned to each MAG

MAG ANNOTATION (276 dereplicated MAGs)



QC / read prep

Assembly / polishing

Binning

MAG QC / taxonomy

Annotation

Abundance

Assembly
strategy:
De Bruijn graph
+ repeat graph
for metagenomes

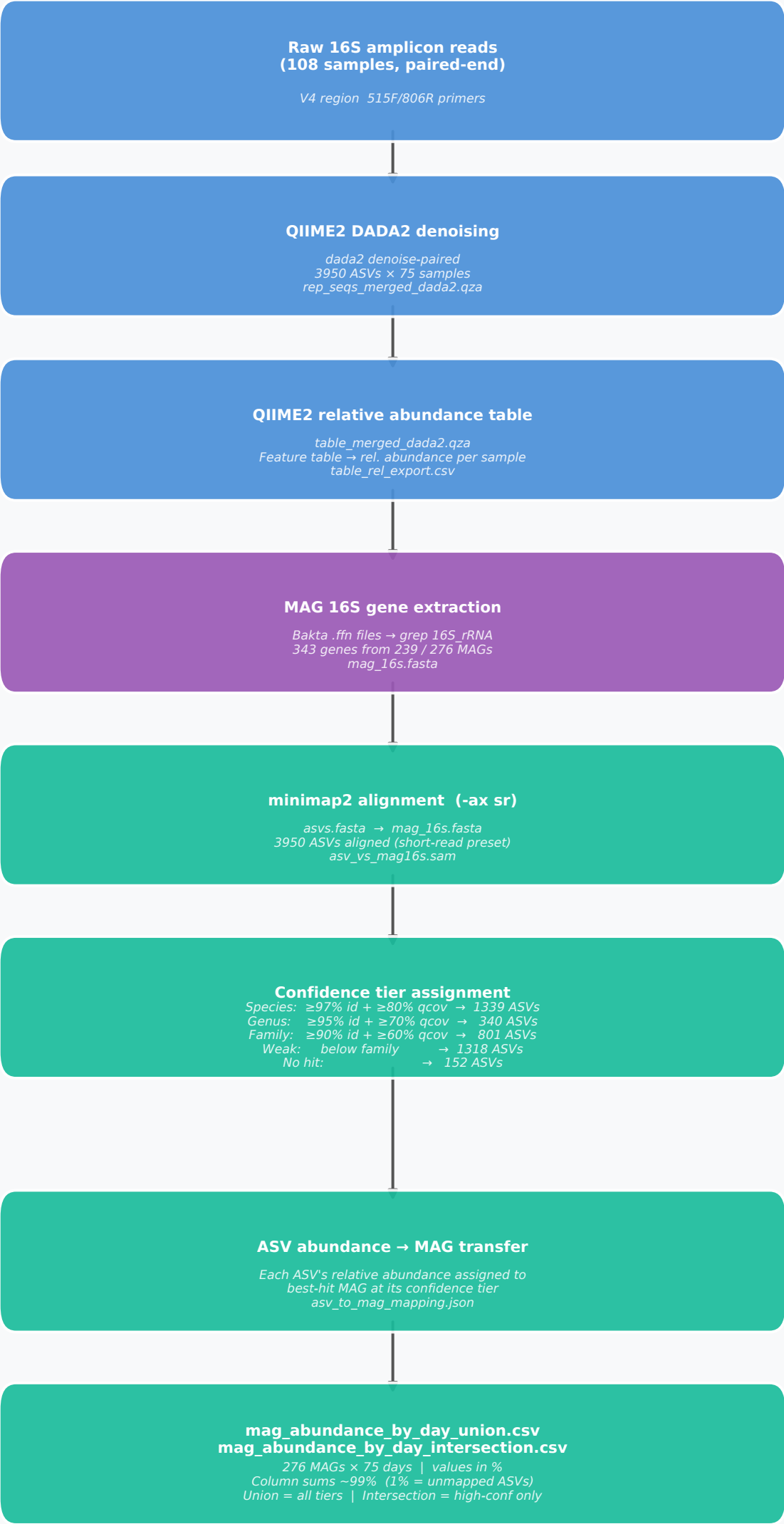
Contigs split
across GPU 0
and GPU 1
(mini_align)

Corrects SNPs
& indels using
per-read
alignments

MAG Abundance Estimation — Two Approaches

Approach A — 16S Amplicon Bridge

75 timepoints (days 1–584) | indirect via DADA2 ASVs



✓ 75 timepoints ✓ dense temporal resolution
△ indirect (2-step error: DADA2 + alignment) △ 37 MAGs have no 16S gene → always zero

Approach B — Direct Metagenomic Mapping

5 timepoints (CAN 1–5) | direct from raw metagenome reads



✓ Direct from metagenome — no 16S intermediary
✓ Captures all 276 MAGs including those without 16S genes
△ Only 5 timepoints | △ 54.5% assembly binned (remainder unmeasured)
→ Recommended use: validate 16S-bridge values at matched timepoints

Annotation Tool Selection — Bakta vs. Alternatives

Justification for using Bakta (full DB) as the primary annotation tool for 276 dereplicated MAGs

	Bakta (chosen)	Prokka	RAST (SEED)	DRAM	Prodigal (ORF only)
Database	RefSeq NR, UniProt SwissProt, ISfinder, CARD, REBASE, AMRFinder, SignalP	UniProt + HAMAP (smaller, older)	SEED subsystems (curated, large)	KEGG, UniRef90, Pfam, CARD, dbCAN, VOG (~500 GB)	None — ORF prediction only
Reference currency	Updated to RefSeq 2024 (curated, versioned)	Databases last updated ~2021; lags RefSeq	SEED updated, but closed-access	Comprehensive but requires large disk & RAM	N/A
16S rRNA recovery	✓ Explicit 16S/23S/5S rRNA annotation → used for ASV-MAG mapping	✓ rRNA annotated but less curated	✓ Included in RASTtk pipeline	✓ Annotates 16S if provided	✗ No rRNA; external tool needed
Reproducibility / local run	✓ Fully local, docker/conda, version-pinned	✓ Local / conda	✗ Web-based API or RASTtk server; queue delays, no versioning	✓ Local conda but heavy setup (500 GB refdata)	✓ Ultrafast, no DB needed
Runtime (×276 MAGs)	~2 h total (16 parallel × 8 t)	Similar to Bakta	Web queue: days RASTtk local: OK but setup complex	Very slow: hours/MAG 276 MAGs = weeks	< 1 min / MAG
Output formats	.gff3 .gbff .tsv .ffn .faa .log (INSDC-compliant)	.gff .gbk .faa .ffn .tsv .txt	.gff .gbk .tsv SEED-specific IDs	.tsv distill report + genome summaries	.gff .faa .fna
Functional depth	Gene function, EC, COG, GO, eggNOG, ARG, IS elements	Gene function + COG (no CARD/IS/AMR by default)	Subsystem-based; good metabolic context	Best metabolism context; KEGG pathway maps	ORF boundaries only; no function
MAG / draft genome support	✓ Designed for MAGs & draft genomes	✓ Widely used for MAGs but showing age	✓ Works; not optimised for MAGs	✓ Designed for MAGs; complex distillation step required	✓ Core of most tools (Bakta uses it internally)

Why Bakta was chosen for this pipeline

- Explicit rRNA annotation (16S / 23S / 5S): Bakta outputs .ffn files containing all non-coding RNA sequences, including annotated 16S rRNA genes. These were used directly to build the MAG–16S reference (mag_16s.fasta) for the ASV → MAG abundance bridge. Prokka also annotates rRNA, but Bakta's 16S loci carry RefSeq-backed gene IDs that make downstream matching more reliable.
- Currency of reference databases: Bakta ships versioned snapshots of RefSeq NR, UniProt Swiss-Prot, ISfinder, CARD, REBASE, and AMRFinderPlus. Prokka's bundled databases were last comprehensively updated in 2021. For a 2026 wastewater microbiome project (dominated by EBPR taxa with many 'hypothetical' proteins), current databases significantly improve hit rates.
- Local, reproducible, version-pinned: RAST requires either a web API (introducing queue latency and non-reproducibility) or a self-hosted RASTtk instance. Bakta runs offline via conda with an explicit DB version – essential for a shared HPC environment where connectivity and API stability cannot be guaranteed.
- DRAM not run (opted out): DRAM offers the richest metabolic reconstruction but requires ~500 GB of reference data, takes hours per MAG (weeks for 276 MAGs serial), and is primarily useful for metabolic pathway distillation – a downstream analysis step. The Bakta + eggNOG combination (KEGG K0, COG, CAZy, GO, EC numbers) provides equivalent functional coverage for the community ecology questions (GAO/PAO enrichment, nutrient cycling) at a fraction of the compute cost.
- INSDC-compliant output: Bakta produces .gbff files that can be deposited directly to NCBI/ENA. This is required for the KBase MAG + genome import workflow already configured in kbase_mag_import.csv and kbase_genome_import.csv.

Better than alternatives Adequate Some limitation Significant limitation